

Sirius Hou

+1-(647)-936-1096 | y45hou@uwaterloo.ca | [Portfolio](#) | [LinkedIn](#) | [GitHub](#)

EDUCATION

University of Waterloo

Ontario, Canada

Candidate for Bachelor of Computer Science + AI Specialization

2021 – 2026

- Cumulative GPA: **3.9/4.0** + Term Distinction x 6, Undergraduate Research Assistant, Teaching Assistant
- UW CS Upper-Year Scholarship + UW President's Scholarship

TECHNICAL SKILLS

Languages: C, C++, C#, Python (PyTorch, NumPy, Pandas), Kotlin, Java, Go, TypeScript, JavaScript, HTML/CSS

Tools: Unreal Engine, Unity, React, Node.js, Qt, Three.js, AWS, PostgreSQL, MongoDB, Android Studio, Git

EXPERIENCE

SideFX Software | 3D Software Developer

May – Dec 2025

- Re-architected and implemented the **Full Body Inverse Kinematics** tool in **Houdini** with **C++ & Python**, supported multi-character editing and configuration auto-restoration, reducing rigging errors by $\sim 40\%$ and cutting manual setup time by $\sim 50\%$.
- Architected **custom node UI** and internal **APIs** for **Houdini Digital Assets**, enabling flexible control shape/color customization for character posing, adopted by **50+** custom tools.
- Engineered the **Effect Tool**, a modular, **user-oriented framework** that allows designers to integrate and execute custom **APEX solver graphs** directly within Houdini without backend coding, cutting design time by $\sim 30\%$.

Christie Digital Systems | Application Software Developer

Sep – Dec 2024

- Developed a **black level uniformity correction** algorithm using **MATLAB**, reduced brightness and color inconsistencies at the edges of projected images on laser projectors, which improved color accuracy by $\sim 30\%$.
- Integrated the algorithm into the calibration pipeline using **C++** and **Qt**, streamlined the calibration process with **automated camera calibration** and **image sampling**, improving calibration accuracy by $\sim 20\%$.
- Upgraded software & web **UI (QT/JS)** for **real-time** RGB laser level adjustments based on user feedback.

Ford Motor Company | Embedded Software Developer

Jan – Apr 2024

- Built **C++**-based telemetry preprocessing services on **AWS**, focusing on scalable data scrubbing and ingestion efficiency, achieving $\sim 15\%$ lower preprocessing latency.
- Designed a **multi-threaded** data ingestion pipeline with **JavaScript** sandbox and **PostgreSQL** to safely execute **1,000+** concurrent intents, reducing vehicle sensor data processing latency by $\sim 23\%$.

Behaviour Interactive Inc. | Software Engineer

May – Aug 2023

- Implemented **Unreal Engine** components using **C++** for **real-time, in-editor actor collision detection** with customizable visual indicators, streamlining level design and boosting team productivity by $\sim 40\%$.
- Optimized a **custom Unreal Engine plugin (Tile Editor)** to automate the conversion of **700+** game scene tiles from blueprints to levels, saving **100+** hours of manual work and greatly improved design efficiency.

University of Waterloo | Research Assistant ([URA](#))

Jul – Dec 2024

- Collaborated with **Prof. Jian Zhao** and designed **VR** scenes in **Unity** with mazes and interactable objects to test **around-ear gestures** for **VR locomotion**, improving **accessibility** for seated and lower-limb disabled users.
- Programmed VR features in **C#**, including **locomotion mechanics**, **humanoid model rigging**, **animation**, and **interactions** with game objects, enhancing the immersive gameplay experience.

PROJECTS

Collaborative Whiteboard Android App ([YADA](#)) | Kotlin, Jetpack Compose

2023

- Developed a **responsive** drawing app supporting **real-time, low-latency** collaboration among multiple users.
- Implemented drawing tools (**pen, eraser, shapes**), with **undo/redo** functionalities for seamless user experience.

WLP4 (Simplified C Language) Compiler | C++

2023

- Developed a full WLP4 compiler in **C++**, featuring maximal-munch lexical analysis, nested-procedure & closure support, and optimization passes (constant folding, in-lining), yielding $\sim 20\%$ faster runtime execution.

UW AI Course Scheduler ([UWCS](#)) | Python, Pandas, BeautifulSoup, OpenAI API

2023

- Built an **LLM-enhanced** intelligent scheduling tool for UW students using OpenAI GPT 3.5 API, automating class data extraction, conflict-free schedule generation, and personalization, adopted by **100+** users.